Shubhranil B

M.S by Research
Center For Machine Intelligence and Data Science (C-MInDS)
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Degree	University	Institute	Year	CPI / %
MS by Research	IIT Bombay	IIT Bombay	2023–26	8.29
B.Tech	MAKAUT	IEM Kolkata	2018–22	8.97
Intermediate/+2	–	DAV Public School	2015–17	90.6

PROFESSIONAL SUMMARY

MS by Research candidate at IIT Bombay (C-MInDS) with a CVPR 2025 publication, specializing in computer vision, 3D scene understanding, and multimodal learning. Experienced in LLM fine-tuning (QLoRA), RAG pipelines, and vision transformers. Seeking roles as an AI/ML Engineer or Data Scientist.

PUBLICATIONS

- Rathore V., **Shubhranil B**, Dutta S., Mehrotra S., Kira Z., Banerjee B. (2025). **When Domain Generalization meets Generalized Category Discovery: An Adaptive Task-Arithmetic Driven Approach**. *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.*

M.S. BY RESEARCH

- **Open Vocabulary 3D Segmentation using MAMBA**
  (M.S. Thesis, Advisor: Prof. Biplab Banerjee, CSRE, IIT Bombay) *(Autumn 2025)*
  - Designed a hybrid **MAMBA-Transformer** architecture which can semantically segment familiar and unseen objects in 3D point clouds.
  - Aligned 3D features with **CLIP text-embeddings** and 2D RGB features to extract semantically-rich point cloud representations, enabling **zero-shot segmentation** of unseen objects.
  - Developed a **semantic-aware scan strategy** to enhance object localization in complex 3D scenes.

- **Domain Generalized Category Discovery**
  (M.S. Thesis, Advisor: Prof. Biplab Banerjee, CSRE, IIT Bombay) *(Autumn 2024)*
  - Proposed **DG^{2}CD-Net**, a novel framework, enabling clustering of known and novel categories in **unseen** target domains without access to target data during training.
  - Achieved **73.30%** accuracy on **PACS** dataset and **53.86%** on **OfficeHome**, representing a **6%** average improvement over standard baselines in the **DG-GCD** setting.
  - Developed an **episodic training** strategy to improve **cross-domain generalization** by introducing **marginal loss** (to better separate known and novel classes), and implementing **domain alignment loss** (to bring same classes closer across different domains).
  - https://github.com/Shubh-Nil/D-GCD

SELF PROJECTS

- **Medical Question Answering with RAG + QLoRA**: *(Spring 2025)*
  - Designed a **retrieval-augmented QA system** using **HuggingFace**, **PubMed** and medical guidelines, to provide **trustworthy answers** grounded with citations.
  - Applied **QLoRA** adapters on **LLama-2-7B** model to improve formatting, and citation correctness.
  - https://github.com/Shubh-Nil/Medical-QA

- **Built ViT + DINO from Scratch**: *(Spring 2025)*
  - Implemented a **Vision Transformer (ViT-B/14)** from scratch with **86 million** parameters, including patch embedding, multi-head self-attention, and transformer encoder blocks.
  - Integrated the **DINO self-distillation** strategy with a student–teacher framework to categorize and understand images **without manual labels**, in self-supervised manner.
  - https://github.com/Shubh-Nil/ViT-DINO

- **Built LLM (GPT-2) from Scratch**: *(Autumn 2024)*
  - Built **GPT-2** model (**124 million** parameters) with **multi-head self-attention**, and pre-training from scratch.
  - Fine-tuned on 2 downstream tasks: **text classification** on the **SMS Spam Collection** dataset, and **instruction fine-tuning** on the **Stanford Alpaca** dataset.
  - https://github.com/Shubh-Nil/Scratch-GPT
- **Built FFNN from Scratch**: *(Autumn 2023)*
  - Developed a **Feedforward Neural Network** from scratch using NumPy only, implementing forward and back-propagation and mini-batch SGD.
  - https://github.com/Shubh-Nil/FFNN

COURSE PROJECTS

- **Band Selection in Hyperspectral Images using Genetic Algorithms**
  (*Advanced Methods in Satellite Image Processing* | *Prof. B. K. Mohan, CSRE, IIT Bombay*) *(Spring 2025)*
  - Optimized a subset of spectral bands, capturing **maximum image content**, converging in **50 generations**.
  - Developed **blended crossover** and **mutation** strategies on the **Indian Pines** and **Salinas** dataset.
  - https://github.com/Shubh-Nil/Band-Selection
- **Removing Motion Blur from Images using Stripformer**
  (*Machine Learning for Remote Sensing* | *Prof. Biplab Banerjee, CSRE, IIT Bombay*) *(Spring 2024)*
  - Restored **sharpness** in motion-blurred images, achieving a **PSNR of 33.06** on the **GoPro** dataset.
  - Implemented **StripFormer**, a deblurring model using intra- and inter-stip tokens to detect region-aware blurs.
  - https://github.com/Shubh-Nil/Deblur-StripFormer
- **Controllable Aerial Scene Generation with InfoGAN**
  (*Machine Learning for Remote Sensing* | *Prof. Biplab Banerjee, CSRE, IIT Bombay*) *(Spring 2024)*
  - Improved **Mutual Information to 2.5** in scene-controlled aerial image generation over a regular **GAN**.
  - Fine-tuned **InfoGAN** on the **PatternNet** dataset, to learn disentangled latent representations.
  - https://github.com/Shubh-Nil/InfoGAN
- **Fine-Grained Image Classification on CUB Dataset using EfficientNet**
  (*Machine Learning for Remote Sensing* | *Prof. Biplab Banerjee, CSRE, IIT Bombay*) *(Spring 2024)*
  - Developed a fine-grained bird species classification model on **CUB** dataset, achieving **75% accuracy**.
  - Trained an **EfficientNet-B7** architecture with **Class Normalization**, outperforming standard **CNN** baselines.
  - https://github.com/Shubh-Nil/Classification-on-CUB
- **Food Calorie Estimation from Images using Deep Learning**
  (*Foundations of Machine Learning* | *Prof. Sunita Sarawagi, CSE, IIT Bombay*) *(Autumn 2023)*
  - Designed a **Calorie Estimation** System, achieving **7.6% mean absolute error** across **19** diverse food categories.
  - Implemented **YOLOv8** for precise food detection and employed **GrabCut** for image segmentation.
  - https://github.com/Shubh-Nil/Food-Calorie-Estimation

TECHNICAL SKILLS

- **Programming & Scripting Languages:** Python, C++, C
- **Tools and Technologies:** PyTorch, HuggingFace, Scikit-Learn, LaTeX, Git, Linux

COURSES (IIT BOMBAY)

CS 725: *Foundations of Machine Learning*
GNR 638: *Machine Learning for Remote Sensing*
EE 601: *Statistical Signal Analysis*
SC 607: *Optimization*
CS 601: *Algorithms and Complexity*
GNR 602: *Advanced Satellite Image Processing*
PS 643: *Introduction to AI, Data, and Policy*
SC 625: *System Theory*

POSITIONS OF RESPONSIBILITY

- *Mentored* a batch of 125 students in the *Machine Learning* course (e-PG Diploma in AI) through the tutorials.
- *Assisted* a batch of 150 students in the *Programming for Data Science* course (e-PG Diploma in AI).

ACHIEVEMENTS

- *Achieved an AIR of **329*** in GATE Computer Science Examination *(2023)*

HOBBIES

- **Reading**: thrillers, manga · **Chess**: problem solving, casual play · **Piano**: film music, scales & arpeggios.